FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

FULL TEST – IV
(Paper-2)

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**A, B, D**	21.	**A, B, C**	41.	**A, B, D**
2.	**B, C**	22.	**A, B, C, D**	42.	**A, C**
3.	**A, B**	23.	**A, C**	43.	**B, C**
4.	**A, B, C, D**	24.	**A, B, C**	44.	**A, B, C, D**
5.	**B, D**	25.	**A, B, C**	45.	**B, C, D**
6.	**A, C, D**	26.	**A, C, D**	46.	**B, C**
7.	**B, D**	27.	**C**	47.	**A, B, C, D**
8.	**A, B, D**	28.	**B, C, D**	48.	**A, B, C**
9.	**A**	29.	**A**	49.	**A**
10.	**A**	30.	**B**	50.	**B**
11.	**C**	31.	**A**	51.	**D**
12.	**A**	32.	**D**	52.	**A**
13.	**2**	33.	**8**	53.	**6**
14.	**1**	34.	**2**	54.	**2**
15.	**2**	35.	**4**	55.	**6**
16.	**8**	36.	**9**	56.	**5**
17.	**7**	37.	**9**	57.	**4**
18.	**3**	38.	**2**	58.	**2**
19.	**7**	39.	**9**	59.	**6**
20.	**2**	40.	**5**	60.	**9**

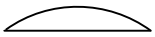
Physics

PART – I

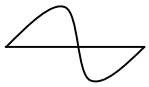
SECTION – A

1. **A, B, D**

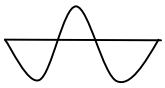
$K = 0.5\pi = \frac{2\pi}{\lambda} \Rightarrow \lambda = 4\text{cm}$



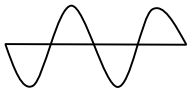
$\Rightarrow L = \frac{\lambda}{2} = 2\text{cm}$



$\Rightarrow L = \lambda = 4\text{cm}$



$\Rightarrow L = \frac{3\lambda}{2} = 6\text{cm}$



$\Rightarrow L = 2\lambda = 8\text{cm}$

2. **B, C**

When the switch 'S' is opened,

$\tau = RC = 1.5\,\text{sec}$

Voltage across the capacitor,

$V_c = 10\left(1 - e^{-t/1.5}\right)$ for $t < T$

The current, $i = \frac{dQ}{dt} = C\ \frac{dV_c}{dt}$

$= \frac{200}{3} \times 10^{-6}\left(e^{-t/1.5}\right)$ for $t < T$

Voltage drop across 100 k μ resister,

$V = iR = \frac{20}{3} e^{-t/1.5}$ for $t < T$

After t = T,

$\tau = 100\ \text{k}\Omega \times 10\ \mu\text{F} = 1\ \text{sec}$

Voltage across capacitor,

$V_c = V_o t^{-\left[(t-T)/\tau\right]} = 10\left(1 - e^{-(T/1.5)}\right), e^{-\left[(t-T)/1\right]}$

The charge on capacitor, $Q = V_c \cdot C$

The current through 100 k μ resistor, $i = -C\frac{dV_c}{dt} = 10^{-3}\left(1 - e^{-T/1.5}\right)\left(1 - e^{-(t-T)/1}\right)$

3. **A, B**

$\Delta\phi = \frac{2\pi}{\lambda}\Delta X$

$\Delta\phi = \frac{2\pi}{16} \times 4 = \frac{\pi}{2}$

$I = 4I_0 \cos^2\frac{\pi}{4} = 2I_0$

4. **A, B, C, D**

Electric field lines kill off on surface of conductor

5. **B, D**

Since maximum current passing through two circuits is same and equal to V/R_1 in (b) and V/R_2 in (c) then $R_1 = R_2$

Current growth equation of L – R circuit is :

$$i = i_0\left(1 - e^{-t/\tau}\right)$$

$$i = i_0\left(1 - e^{\frac{-tR}{L}}\right)$$

Now, $\frac{di}{dt} = \frac{i_0 R}{L} e^{-tR/L}$

At t = 0 slope of I vs t curve:

$$\frac{di}{dt} = \frac{i_0 R}{L}$$

From graph (a) : slope of curve (b) > slope of curve (c)

$$\frac{R_1}{L_1} > \frac{R_2}{L_2}$$

Since, $R_1 = R_2$

$L_1 < L_2$

6. **A, C, D**

Momentum conservation.

$$MV_0 = 2MV$$

$$V = \frac{V_0}{3}$$

Angular momentum conservation about centre of mass,

$$\frac{1}{2}MR^2 \cdot \omega_0 - \frac{4}{3}MV_0R = \left[\frac{MR^2}{2} + M\left(\frac{4}{3}R\right)^2 + \frac{1}{2}(2M)R^2 + 2M\left(\frac{2}{3}R^2\right)\right]\omega$$

$$\omega = \frac{3}{25}\omega_0 - \frac{8}{25}\frac{V_0}{R}$$

For $\omega = 0$

$$\omega_0 = \frac{8V_0}{3R}$$

The energy loss,

$$\Delta E = E_i - E_f$$

$$= \frac{1}{2}MV_0^2 + \frac{1}{2}\left(\frac{MR^2}{2}\right)\cdot\omega^2 - \frac{1}{2}(3M)V^2 = \frac{19}{9}MV_0^2$$

7. **B, D**

$$V_1 = \sqrt{\frac{T}{\mu_1}} = 10\sqrt{2}$$

$$V_2 = \sqrt{\frac{T}{\mu_2}} = \frac{10\sqrt{2}}{\sqrt{3}}$$

Reflection coefficient

$$R = \frac{V_2 - V_1}{V_1 + V_2} = 2 - \sqrt{3}$$

Transmission coefficient,

$$T = \frac{2V_2}{V_1 + V_2} = \sqrt{3} - 1$$

8. **A, B, D**

Initially,

$P_{gas} = P_0A + mg = 112.5 \text{ kPa}$

$w = \int_0^x P_{gas}A \, dx$

$= \int_0^x (P_0A + mg + kx)dx = 56.5 \text{ J}$

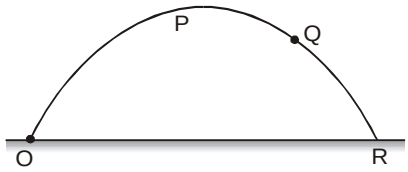
$Q = m\Delta u + w = 118 \text{ J}$

9. **A**

Time of flight of A is 4 seconds which is same as the time of flight if wall was not there.

Time taken by B to touch the inclined roof is 1 second.

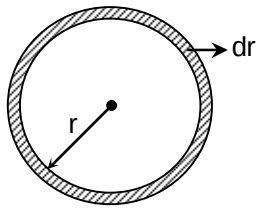
Hence, $S = 20t - \frac{1}{2}gt^2 = 20 \times 1 - 5 \times 1^2 = 15 \text{ m}$



10. **A**

Vertical component = $u\sin\theta = 20 \text{ m/s}$

11. **C**



$dm = (2\pi r dr)\sigma_0\left(1 + \frac{r}{R}\right) = 2\pi\sigma_0\left(r + \frac{r^2}{R}\right)dr$

$M = \int_0^R dm = 2\pi\sigma_0\left(\frac{R^2}{2} + \frac{R^2}{3}\right) = \frac{5\pi\sigma_0R^3}{3}$

$dI = dmr^2 = 2\pi\sigma_0\int_0^R\left(r^3 + \frac{r^4}{R}\right)dr = 2\pi\sigma_0\left(\frac{R^4}{4} + \frac{R^4}{5}\right) = \frac{9}{10}\pi\sigma_0R^4$

$d\tau = g\mu r dm = 2\mu\pi\sigma_0 g\left(r^2 + \frac{r^3}{R}\right)dr$

$\tau = 2\pi\sigma_0\mu g\int_0^R\left(r^2 + \frac{r^3}{R}\right)dr = 2\pi\sigma_0\left(\frac{R^3}{3} + \frac{R^3}{4}\right)\mu g = \frac{7}{6}\pi\mu\sigma_0 gR^3$

$\tau = I\alpha \Rightarrow \frac{7}{6}\pi\mu\sigma_0 gR^3 = \frac{9}{10}\pi\sigma_0R^4\alpha$

$\alpha = \frac{35\mu g}{27R}$

12. **A**

$\omega = \omega_0 - \alpha t$

$dE = (\mu dmgr)(\omega_0 - \alpha t)dt = dmCdT$

$\Rightarrow \frac{dT}{dt} = \frac{\mu gr}{C}(\omega_0 - \alpha t)$

SECTION – C

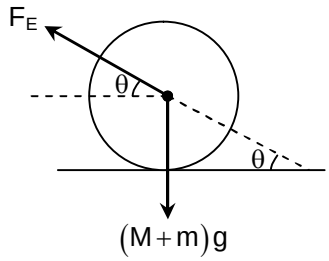
13. **2**

$(M+m)g > F_E \sin\theta$

$> \dfrac{27Qq}{256\pi\varepsilon_0 R^2}$

$(M+m)g > 3mg$

$M > 2m \Rightarrow \eta = 2$



14. **1**

$P_0V_0 = nRT_0$

$W_1 = P_0(4V_0 - V_0) = 3nRT_0$

$W_2 = mRT_0 \ln 4 = 2nRT_0 \ln 2$

$\dfrac{W_1}{W_2} = \dfrac{3nRT_0}{2nRT_0 \ln 2} = \dfrac{3}{2\ln 2} \Rightarrow \alpha = \dfrac{3}{2} \Rightarrow \dfrac{2}{3}\alpha = 1$

15. **2**

$1.67\times10^{-27} v_0 = \dfrac{6.62\times10^{-34}}{61\times10^{-9}} \Rightarrow v_0 = 6.5 \text{ m/s} \Rightarrow n = 2$

16. **8**

$\dfrac{1}{2}L(4) = 16 \Rightarrow$ L = 8 H

4R = 320 $\Rightarrow$ R = 80Ω

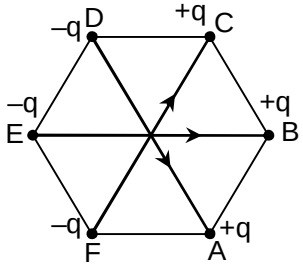
When connected to the battery, R_{net} = 100Ω , L = 8 H

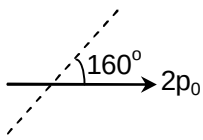
$\tau = \dfrac{8}{100} \Rightarrow 100\tau = 8$

17. **7**

$E = \dfrac{K(2P_0)}{x^3}\sqrt{1+\dfrac{3}{4}} = \dfrac{1}{4\pi\varepsilon_0}\dfrac{4aq}{x^3}\dfrac{\sqrt{7}}{2} = \dfrac{\sqrt{7}aq}{2\pi\varepsilon_0 x^3}$

$\Rightarrow \alpha = 7$





18. **3**

$$E = \frac{1}{2}\left(\frac{C}{2}\right)V^2 \qquad Q_i = \frac{CV}{2} \qquad Q_f = \frac{2CV}{3}$$

$$Q_f - Q_i = \frac{2CV}{3} - \frac{CV}{2} = \frac{CV}{6}$$

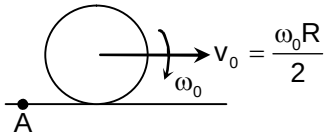
$$\frac{E}{W} = \frac{CV^2/4}{CV^2/6} = \frac{3}{2} \Rightarrow \frac{2E}{W} = 3$$

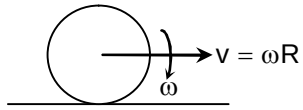
19. **7**

$$\frac{m\omega_0 R^2}{2} + \frac{mR^2}{2}\omega_0 = m\omega_0 R^2 = m\omega R^2 + \frac{mR^2}{2}\omega$$

$$\Rightarrow \omega_0 = \frac{3}{2}\omega \Rightarrow \omega = \frac{2\omega_0}{3}$$

$$\Rightarrow v = \omega R = \frac{2}{3}\omega_0 R = \frac{4v_0}{3}$$





$$\Rightarrow \frac{16v_0^2}{9} = v_0^2 + 2\mu g x$$

$$2\mu g x = \frac{7v_0^2}{9} \Rightarrow x = \frac{7v_0^2}{18\mu g} = \frac{7\omega_0^2 R^2}{72\mu g} = 7 \text{ m}$$

20. **2**

$$K_0 = \frac{1}{2}\alpha x_0^2$$

Kinetic energy just after 3rd collision = $\left(\frac{1}{2}\right)^6 K_0 = \frac{1}{64}K_0$

$$\frac{1}{64}\left(\frac{1}{2}\alpha x_0^2\right) = \frac{1}{2}\alpha x^2$$

$$\Rightarrow x^2 = \frac{x_0^2}{64} \Rightarrow x = \frac{x_0}{8}$$

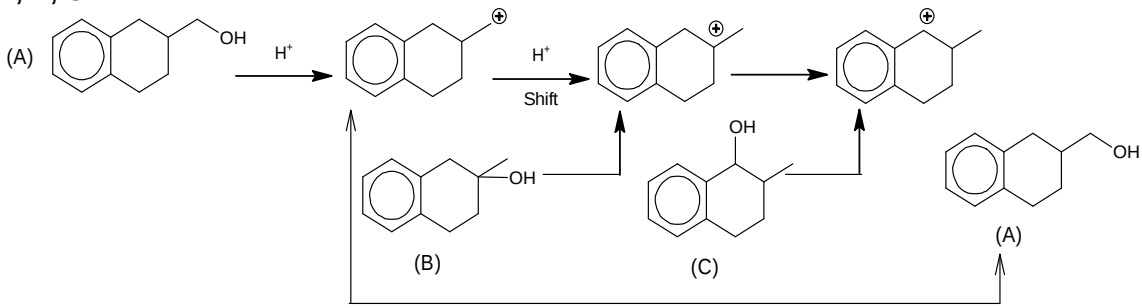
$$F = \alpha x = \frac{\alpha x_0}{8} = 2 \text{ N}$$

Chemistry

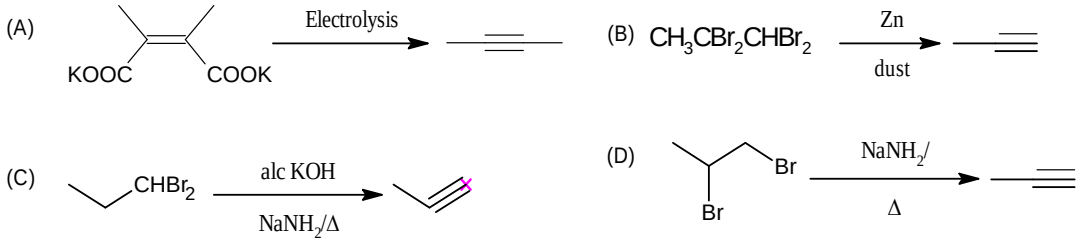
PART – II

SECTION – A

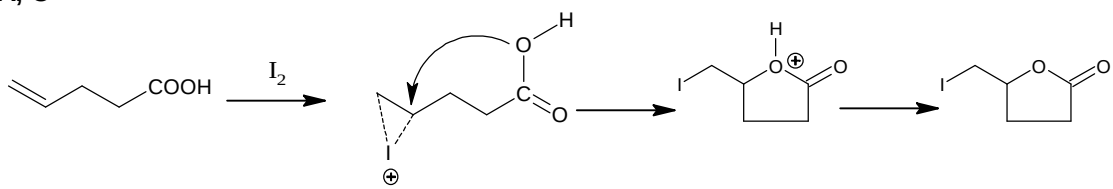
21. **A, B, C**



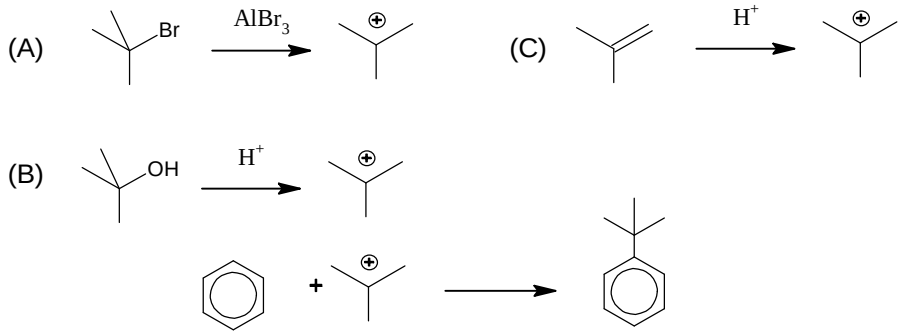
22. **A, B, C, D**



23. **A, C**

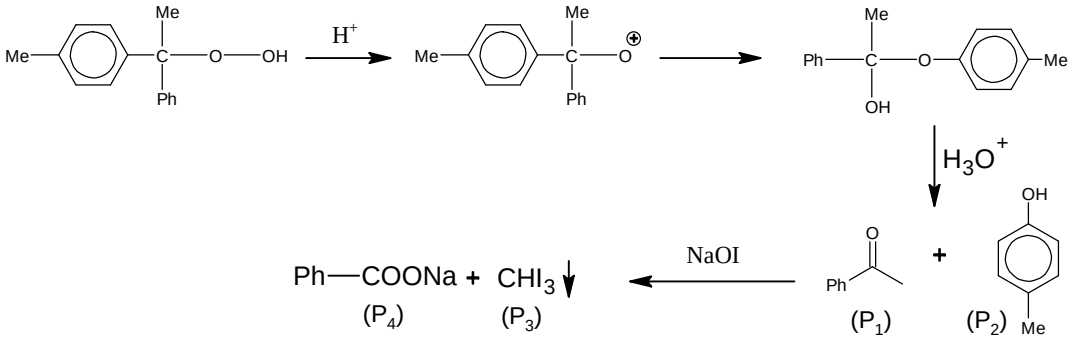


24. **A, B, C**



25. **A, B, C**
Presence of strong electron withdrawing group inhabitants the fried craft alkylation.

26. **A, C, D**



$CHI_3 + Ag \longrightarrow HC\equiv CH$

$Ph—COONa \xrightarrow{\text{sodalime}} Ph—H$

27. **C**
Fact

28. **B, C, D**
$Br_2\,(ON = 0)$ reduces to $Br^-\,(ON = -1)$ and oxidises to $BrO_3^-\,(ON = +5)$

29. **A**
Meq. Of KI ≡ meq. of $CuSO_4$ ≡ meq. of I_2 ≡ meq. of hypo
$0.4\times10^3\times1 \qquad 50\times0.2\times1$
$= 40 \qquad = 10$ (limiting reagent)
$\therefore$ meq. of hypo $= 10 = M\times x\times V_{mL}$
$= 0.4\times2\times V_{mL} = 12.5mL$

30. **B**
meq. of hypo ≡ meq. of Cu^{2+} ≡ meq. of Cu
$12.12 \times 0.1 = \dfrac{w_{Cu}}{63.5/2}\times1000 \Rightarrow w_{Cu} = 0.076g$
$\therefore \quad \%\text{ purity} = \dfrac{0.076}{1.1}\times1000 = 6.99 \approx 7\%$

31. **A**
The rate of the reaction = Rate = K $[A_0]$
= 0.0462 × 100 = 4.62 bacterias per minute

32. **D**
$t_{1/2} = \dfrac{0693}{K} = \dfrac{0.693}{0.0462} = 15\,min$

SECTION – C

33. **8**
Fact

34. **2**

Fact.

35. **4**

Volume of the unit cell, $a^3 = \left(5\times10^{-8}\right)^3$

$= 1.25\times10^{-22}\text{cm}^3 \qquad (\because\ a = 5\overset{o}{A})$

Density (ρ) = 4.0 g cm^{-3}

Mass of unit cell $= V\times\rho = 1.25\times10^{-22}\times4.0$

$= 5.0\times10^{-22}\text{g}$

Number of FeO molecules per unit cell $= \dfrac{5.0\times10^{-22}\text{g}}{1.195\times10^{-22}\text{g}}$

$= 4.19 \approx 4$

Thus, there are four Fe^{2+} ions and four O^{2-} ions in each unit cell.

36. **9**

In conversion of $^{238}_{92}U$ to $^{206}_{82}Pb, 8\alpha$-particles and 6β particles are ejected.

The number of gaseous moles initially = 1 mole

The number of gaseous moles finally = 1 + 8 moles; (1 mole from air and 8 mole of $_2He^4$)

So the ratio = 9/1 = 9

37. **9**

For AgCl, molarity = normality

$\kappa = 3.41\times10^{-6} - 1.60\times10^{-6} = 1.81\times10^{-6}\,\Omega^{-1}\text{cm}^{-1}$

For saturated solution of sparingly soluble salt,

$\lambda_V = \lambda^\infty$ and solubility = normality

$\therefore \qquad \lambda_V = \lambda^\infty = \dfrac{1000\times\kappa}{s}$

$\therefore \qquad s = \dfrac{1000\times\kappa}{\lambda^\infty} = \dfrac{1000\times1.81\times10^{-6}}{138.3} = 1.31\times10^{-5}$

Now,

$K_{sp(AgCl)} = s^2 = \left(1.31\times10^{-5}\right)^2 = 0.1716\times10^{-9}$

$\therefore \qquad x = 9$

38. **2**

Considering the reduction reaction

$$A^{(x+n)+} + ne \longrightarrow A^{x+}$$

Oxidized form Reduced form

$\because \qquad [A^{x+}] = 24.4 \qquad \therefore \qquad [A^{(x+n)+}] = 75.6,\ E_{RP} = 0.101\text{V}$

Now from Nernst's equation

$$E_{RP} = E^0_{RP} + \frac{0.059}{n}\log\frac{[OF]}{[RF]}$$

$$0.101 = E^0_{RP} + \frac{0.059}{n}\log\frac{75.6}{24.4} \quad \text{....(i)}$$

If $[A^{x+}] = 48.8$; $[A^{(x+n)+}] = 51.2$, $E = 0.115$

$$\therefore \quad 0.115 = E^0_{RP} + \frac{0.059}{n}\log\frac{51.2}{48.8} \quad \text{...(ii)}$$

By eqs. (ii) – (i),

$$\therefore \quad 0.0014 = \frac{0.059}{n}\left[\log\frac{51.2}{48.8} - \log\frac{75.6}{24.4}\right]$$

$$\therefore \quad n = \frac{0.059}{0.014}[0.49 - 0.020]$$

$$n = 1.98 \approx 2$$

39. **9**

In Layer M

$n_A = x \quad n_B = y$

$$\frac{x}{x+y} = 0.2 \Rightarrow \frac{y}{x} + 1 = 5 \Rightarrow \frac{y}{x} = 4 \Rightarrow y = 4x$$

In layer N

$$\frac{(i-x)}{(1-x)(3-y)} = 0.6$$

$$\frac{3-y}{1-x} + 1 = \frac{1}{0.6} \Rightarrow \frac{3-y}{1-x} = \frac{4}{6} = \frac{2}{3}$$

$$9 - 3y = 2 - 2x$$

$$9 - 12x = 2 - 2x$$

$$7 = 10x \Rightarrow x = 0.7, y = 2.8$$

$$w_M = 0.7 \times 20 + 0.7 \times 4 \times 40 = 14 + 7 \times 16 = 126$$

$$w_N = 0.3 \times 20 + 0.2 \times 40 = 14$$

$$w_M / w_N = 9$$

40. **5**

	mass	mole	moler ratio
C	6w	$\frac{6w}{12}$	0.5=3
H	w	$\frac{w}{1}$	1 = 6
N	$\frac{7w}{1.5}$	$\frac{7w}{1.5\times14}$	$\frac{1}{3} = 2$

M.wt.$\left[(C_3H_6N_2)_n > 100 \Rightarrow n = 2\right]$

Mathematics

PART – III

SECTION – A

41. **A, B, D**

$$\frac{a}{1+\alpha^2\beta^2}=\frac{b}{\alpha^2+\beta^2}=\frac{c}{\left(1-\alpha^2\right)\left(1+\beta^2\right)}=\frac{2s}{2+2\beta^2}$$

$$\Rightarrow \frac{s}{1+\beta^2}=\frac{a}{1+\alpha^2\beta^2}\Rightarrow \frac{s-a}{\beta^2\left(1-\alpha^2\right)}=\frac{a}{1+\alpha^2\beta^2}$$

$$\Rightarrow s-a=\frac{a\left(1-\alpha^2\right)\beta^2}{1+\alpha^2\beta^2}$$

similarly $s - b = \dfrac{b\left(1-\alpha^2\right)}{\alpha^2+\beta^2}$

$$s-c=\frac{c\alpha^2}{1-\alpha^2}\ \&\ s=\frac{b\left(1+\beta^2\right)}{\alpha^2+\beta^2}$$

$$\therefore \tan^2(A/2)=\frac{(s-b)(s-c)}{s(s-a)}=\frac{\dfrac{b\left(1-\alpha^2\right)}{\alpha^2+\beta^2}\cdot\dfrac{c\alpha^2}{1-\alpha^2}}{\dfrac{b\left(1+\beta^2\right)}{\alpha^2+\beta^2}\dfrac{a\left(1-\alpha^2\right)\beta^2}{1+\alpha^2\beta^2}}$$

$$\left[\text{Q } a=\frac{c\left(1+\alpha^2\beta^2\right)}{\left(1-\alpha^2\right)\left(1+\beta^2\right)}\right]$$

$$=\frac{\alpha^2}{\beta^2}$$

$$A=2\tan^{-1}(\alpha/\beta)$$

$$B=2\tan^{-1}(\alpha\beta)$$

$$\therefore \text{Reqd area} = \frac{1}{2}bc\sin A = \frac{1}{2}bc.\frac{2\tan(A/2)}{1+\tan^2(A/2)}=\frac{bc\alpha\beta}{\alpha^2+\beta^2}$$

42. **A, C**

$$a_2=a_1+\frac{3}{2}b_1=48,\ b_2=\frac{b_1}{2}=16$$

$$a_3=48+\frac{3}{2}\times 16=72,\ b_3=\frac{16}{2}=8$$

So, the three loops from i = 1 to i = 3 are alike Now, area of ith loop (square) = $\frac{1}{2}\left(\text{diagonal}\right)^2$

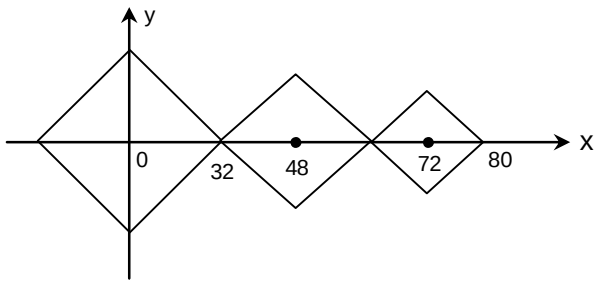
$A_i = \frac{1}{2}(2b_i)^2 = 2b_i^2$

$\therefore \frac{A_{i+1}}{A_i} = \frac{2(b_{i+1})^2}{2(b_i)^2} = \frac{1}{4}$

So, the areas form a G.P series

$\therefore$ sum of the G.P up to infinite term is

$\frac{A_i}{1-r} = 2(32)^2 \times 1\frac{1}{1-\frac{1}{4}} = \frac{8}{3}(32)^2$ sq. units



43. **B, C**

$|\bar{a}| = 1, |\bar{b}| = 4, \bar{a}.\bar{b} = 2 \;\&\; \bar{c} + 3\bar{b} = 2\bar{a} \times \bar{b}$

$Q\ \bar{a}.\bar{b} = 2$

$\Rightarrow |\bar{a}||\bar{b}|\cos\theta = 2$

$\Rightarrow \cos\theta = \frac{2}{|\bar{a}||\bar{b}|} = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$

Again $|\bar{c} + 3\bar{b}|^2 = |2\bar{a} \times \bar{b}|^2$

$\Rightarrow |\bar{c}|^2 + 9|\bar{b}|^2 + 2\bar{c}.3\bar{b} = 4|\bar{a}|^2 |\bar{b}|^2 \sin^2\theta$

$\Rightarrow |\bar{c}|^2 + 144 + 6\bar{b}.\bar{c} = 48$

Also, $\bar{c} = 2\bar{a} \times \bar{b} - 3\bar{b}$

$\Rightarrow \bar{b}.\bar{c} = \bar{b}.(2\bar{a} \times \bar{b}) - 3|\bar{b}|^2$

$\Rightarrow \bar{b}.\bar{c} = 0 - 3 \times 16 = -48$

$\therefore |\bar{c}|^2 + 96 - 6 \times 48 = 0$

$\Rightarrow |\bar{c}|^2 = 192$

$\Rightarrow |\bar{c}| = 8\sqrt{3}$

$\bar{b}.\bar{c} = |\bar{b}||\bar{c}|\cos\alpha \Rightarrow -48 = 4 \times 8\sqrt{3}\cos\alpha$

$\Rightarrow \cos\alpha = -\frac{6}{4\sqrt{3}} = -\frac{\sqrt{3}}{2}$

$\Rightarrow \alpha = \frac{5\pi}{6}$

44. **A, B, C, D**

1 + x is never zero, so 1 + f(x) is never zero. It is 1 for x = 0, so it is always positive. Hence $f''(x)$ is always positive $f'(0) = 0$ so $f'(x) > 0,\ \forall x > 0$

Hence so in particular, $1 + f(x) \geq 2,\ \forall x$

$f''(x) \leq \frac{1+x}{2}$

Integrating $f'(x) \le f'(0) + \frac{x}{2} + \frac{x^2}{4} = \frac{x}{2} + \frac{x^2}{4}$

Again integrating

$f(x) \le f(0) + \frac{x^2}{4} + \frac{x^3}{12}$

Hence $f(1) \le 1 + \frac{1}{4} + \frac{1}{12} = \frac{4}{3}$

45. **B, C, D**

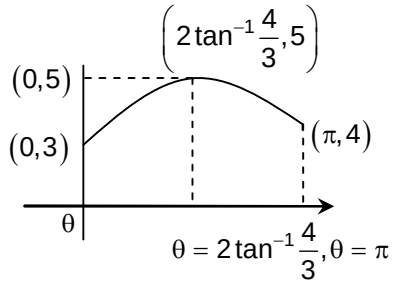
Let angle between $\bar{a}$ & $\bar{b}$ be $\theta \left|\bar{a}\right| = \left|\bar{b}\right| = 1, \frac{\theta}{2}$

$\left|\bar{a} + \bar{b}\right| = 2\cos\frac{\theta}{2}$

$\left|\bar{a} - \bar{b}\right| = 2\sin\frac{\theta}{2}$

$F(\theta) = \frac{3}{2}\left(2\cos\frac{\theta}{2}\right) + 2\left(2\sin\frac{\theta}{2}\right)$

$= 3\cos\frac{\theta}{2} + 4\sin\frac{\theta}{2}, \theta \in [0, \pi]$



$F'(\theta) = -\frac{3}{2}\sin\frac{\theta}{2} + 2\cos\frac{\theta}{2}$

$F'(\theta) = 0 \Rightarrow \tan\left(\frac{\theta}{2}\right) = \frac{4}{3}$

$\therefore F(0) = 3$

$F\left(2\tan^{-1}\frac{4}{3}\right) = 3\left(\frac{3}{5}\right) + 4\left(\frac{4}{5}\right) = 5$

$F(\pi) = 4 \therefore$ Range $[3,5]$

Hence possible integer (s) in the range of $F(\theta)$ in $[0,\pi]$ are 3

46. **B, C**

$f(k) = 1 + \frac{1}{2} + \frac{1}{3} + ... + \frac{1}{k}$

$f(1) = 1$

$f(2) = 1 + \frac{1}{2}$

$f(2017) = 1 + \frac{1}{2} + \frac{1}{3} + ... \frac{1}{2017}$ on adding

$= 1 \times 2017 + \frac{1}{2} \times 2016 + \frac{1}{3} \times 2015 + ... + \frac{1}{2017} \times 1$

$= 1 \times 2017 + \frac{1}{2}(2017 - 1) + \frac{1}{3}(2017 - 2) + ... + \frac{1}{2017}(2017 - 2016)$

$= 2017\left(1 + \frac{1}{2} + \frac{1}{3} + ... + \frac{1}{2017}\right) - \left(\frac{1}{2} + \frac{2}{3} + ... + \frac{2016}{2017}\right)$

$= 2017 f(2017) - \left\{1 - \frac{1}{2} + 1 - \frac{1}{3} + 1 - \frac{1}{4} + ... + 1 - \frac{1}{2017}\right\}$

$= 2017f(2017) - 1\times 2016 + \left(1 + \frac{1}{2} + ... + \frac{1}{2017}\right)$

$= 2017f(2017) - 2016 + f(2017)$

$= 2018f(2017) - 2017$

$a = 2018,\ b = 2017,\ c = -2017$

47. **A, B, C, D**

$x^2 - x\left(\frac{1+\sin^2\theta}{1+\sin\theta}\right) + \frac{\sin\theta - \sin^2\theta}{1+\sin\theta} = 0$

$x^2 - x\left(\sin\theta + \frac{1-\sin\theta}{1+\sin\theta}\right) + \sin\theta\left(\frac{1-\sin\theta}{1+\sin\theta}\right) = 0$

$\Rightarrow x = \sin\theta, \frac{1-\sin\theta}{1+\sin\theta}$

$\Rightarrow \alpha(\theta) = \sin\theta, \beta(\theta) = \frac{1-\sin\theta}{1+\sin\theta}$

Hence the results follows.

48. **A, B, C**

D(x) can at most be polynomial of degree 2

$D(x) = ax^2 + bx + c$

$D(x) = x^2 + x + 1$

$D(1) = 3$

$D(-2) = 4 - 2 + 1 = 3$

$D(-3) = 9 - 3 + 1 = 7$

49. **A**

50. **B**

$r_1 = b - x_1 = OP = OA$

$AP_1 = a/2$

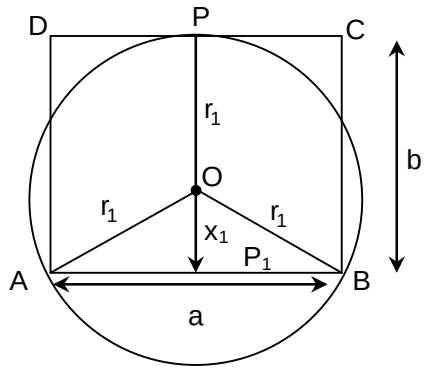
$r_1^2 = x_1^2 + \left(\frac{a}{2}\right)^2 = (b - x_1)^2$

$x_1^2 + \frac{a^2}{4} = b^2 + x_1^2 - 2bx_1$

$x_1 = \frac{4b^2 - a^2}{8b}$

$r_1 = b - x_1 = \frac{4b^2 + a^2}{8b}$

similarly $r_2 = \frac{4a^2 + b^2}{8a}$



$$r_1 + r_2 = \frac{a^3 + b^3 + 4ab(a+b)}{8\ ab}$$

$$= \frac{(a+b)\left(a^2 + 3ab + b^2\right)}{8\ ab}$$

$$= \left(\frac{a+b}{8}\right)\frac{\left[(a-b)^2 + 5ab\right]}{ab}$$

But $(a-b)^2 \geq 0$

$$r_1 + r_2 \geq \frac{(a+b)}{8}.\frac{5ab}{ab}$$

$$\Rightarrow r_1 + r_2 \geq \frac{5(a+b)}{8}$$

51. **D**

52. **A**

$$f'(0) = \lim_{h\to 0}\frac{f(h)-1}{h} = \ln a$$

$$f'(x) = \lim_{h\to 0}\frac{f(x)}{2}\left[\frac{f(h)-1}{h} + \frac{f(-h)-1}{-h}\right] = \frac{f(x)}{2}\ 2\ln a$$

$$\frac{f'(x)}{f(x)} = \ln a\ \Rightarrow \ln f(x) = \ln a^x + C$$

$$f(0) = 1 \Rightarrow C = 0 \Rightarrow f(x) = a^x$$

$$\frac{dy}{dx} = \frac{(x+y)^2}{(x+2)(y-2)}\ \ (\text{put } x+2 = X,\ y-2 = Y)$$

$$\frac{dY}{dX} = \frac{(X+Y)^2}{XY}$$

$$\Rightarrow \frac{y-2}{x+2} - \frac{1}{2}\ln\left|1 + \frac{2(y-2)}{x+2}\right| = 2\ \ln\ (x+2) + C$$

SECTION – C

53. **6**

$$Q(x) = \left(x^3 - 2x^2 + x - 1\right)(x+1) = 0 \Rightarrow \delta = -1$$

$\alpha + \beta + \gamma = 2$, $\alpha\beta+\beta\gamma+\gamma\alpha=1$

$$p(x) = \left(x^2 + 1\right)Q(x) + x^2 - x + 1$$

$$p(\alpha) + p(\beta) + p(\gamma) + p(-1) = \alpha^2 + \beta^2 + \gamma^2 + 1 - (\alpha + \beta + \gamma - 1) + 4$$

54. **2**

Let $\left[\log_{10} 4x\right] = k = \left[\log_{10} x\right]$ then $x \in \left[10^k, \frac{10^{k+1}}{4}\right]$, for k > 0 & k = 0 $\left(10^k, \frac{10^{k+1}}{4}\right)$ is beyond (0, 1) but for k < 0, the whole interval is in (0, 1).

So favourable x is obtained for $k = -1,-2,-3,.....$

for $k = -1, x \in \left[\frac{1}{10},\frac{1}{4}\right)$

for $k = -2, x \in \left[\frac{1}{100},\frac{1}{40}\right)$.....

sum of lenght of these intervals

$$= \left(\frac{1}{4}-\frac{1}{10}\right)+\left(\frac{1}{40}-\frac{1}{100}\right)+.....$$

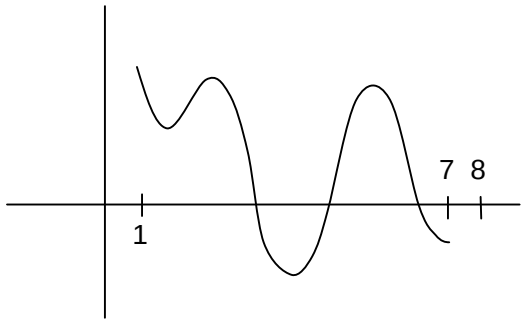
$$= \frac{3}{20}\left(1+\frac{1}{10}+\frac{1}{100}+....\right)=\frac{1}{6}$$

Required probability $= \frac{\frac{1}{6}}{1-0}=\frac{1}{6}$

55. **6**

$f'(x) = f'(x)\left(f(x)\right)^2$

Either f'(x) = 0 at least 4 roots
f(x) = 1 at least 5 roots
f(x) = - 1 at least 2 roots
at least 11 roots.



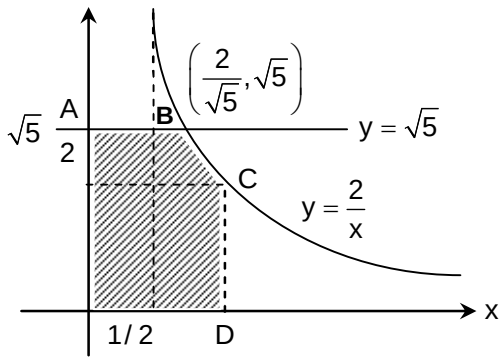
56. **5**

f(x) will be below $y = \sqrt{5}$ & $y = \frac{2}{x}$

$$\therefore I \le \int_0^{2/\sqrt{5}} \sqrt{5}dx + \int_{\frac{2}{\sqrt{5}}}^{1} \frac{2}{x}dx$$

$$= \sqrt{5}\left(\frac{2}{\sqrt{5}}\right)+2\ln 1\ -2\ \ln\left(\frac{2}{\sqrt{5}}\right)$$

$$= 2+2\ln\left(\frac{\sqrt{5}}{2}\right)$$



57. **4**

There are ${}^{40}C_4$ ways of drawing 4 slips ${}^{10}C_2$ ways to choose a & b. After a & b have been chosen there are ${}^4C_2$ ways of choosing 2 slips labelled a & ${}^4C_2$ ways of choosing 2 slips labelled b

$\therefore$ Required Probability $= \frac{{}^{10}C_2 \times {}^4C_2 \times {}^4C_2}{{}^{40}C_4} = \frac{36.{}^{10}C_2}{{}^{40}C_4}$

58. **2**

Since

$$\left(\int_a^b f(x)g(x)dx\right)^2 \le \int_a^b f^2(x)dx\int_a^b g^2(x)dx$$

$$\therefore \int_a^b xf(x)f'(x)dx = \frac{1}{2}\int_a^b x\left(f^2(x)\right)'dx$$

$$= \frac{1}{2}\left[\left[xf^2(x)\right]_a^b - \int_a^b f^2(x)dx\right] = \frac{1}{2}$$

Hence by Schwartz's inequality

$$\frac{1}{4} \le \int_a^b \left(f'(x)\right)^2 dx.\int_a^b x^2f^2(x)dx$$

59. **6**

$$\sin\pi x + 1 = \left|\ln|x|^3\right| + 1$$

$$\sin\pi x = \left|\left(\ln|x|\right)^3\right|$$

No of solutions 6

60. **9**

A_n denotes no. of n digit sequence that end in 0 or 4
B_n denotes no. of n digit sequence that end in 1 or 3
C_n denotes no. of n digit sequence that end in 2
$A_{n+1} = B_n$ ($\because$ each sequence in A_{n+1}can be converted into a sequence in B_n by deleting its last digit). $B_{n+1}= A_n + 2\ C_n$ ($\because$ each sequence in A_n can be converted into a sequence in B_{n+1} by adding a 1 (if it ends with 0) or a 3 (if it ends with 4) to its end, 8 each sequence in C_n can be converted into a sequence in B_{n+1} by adding a 1 or a 3 to it. $C_{n+1} = B_n$ (by deleting 2 at its end)
$B_{n+1} = 3B_{n-1}$ for $n \ge 2$, $B_1 = 2$, $B_2 = 4$
$\therefore \quad A_{10} + B_{10} + C_{10} = 2B_9 + B_{10} = 4.3^4 + 4.3^4 = 648$